

InsureScope - Insurance Market Research and Predictive Analytics Platform

Consolidated End-to-End Business Analysis Document

Prepared by	Business Analyst
Version	1.0
Status	Final
Completion date	xx Feb 20xx
Confidentiality	Confidential - Portfolio and professional review
Data classification	All insurance, quotation, policy and customer data are fictional
Product status	Approved release candidate with documented limitations

This document consolidates the approved business analysis, implementation evidence, testing records, governance decisions and release acceptance for InsureScope. It must not be represented as production-ready insurance software.

1. Document Control

Version	Status	Description	Prepared by	Date
1.0	Final	Consolidated end-to-end Business Analysis document	Business Analyst	xx Feb 20xx

1.1 Review and approval

Review area	Owner	Status	Evidence
Business and release decision	Technical Business Analyst	Approved	BA-ML-31 final release decision
Market Research MVP release boundary	Technical Business Analyst	Approved	Release acceptance record
Predictive-ML generation and model activity	Technical Business Analyst	Not authorised	Continuing governance prohibitions
Public deployment	Not assigned	Not completed	No configured deployment workflow

1.2 Document ownership

The document is owned by the Business Analyst function. It consolidates business, technical, data, API, testing and predictive-ML governance evidence without attributing software engineering or data-science ownership to the Technical Business Analyst.

1.3 Source-document inventory

ID	Source	Use	Status
SRC-01	README.md	Product purpose, categories, demonstration roles, technology and MVP exclusions	Current
SRC-02	BA-ML-31	Final release acceptance, API inventory, testing, performance, limitations and approval	Approved
SRC-03	BA-ML-27	Dataset V3 generation implementation contract	Approved; execution blocked
SRC-04	BA-ML-28	Challenge confidentiality amendment	Approved governance decision
SRC-05	BA-ML-29	Feature schema and temporary generation contract	Approved
SRC-06	BA-ML-30	Preprocessing approval and implementation evidence	Approved temporary work only
SRC-07	Database design and loading records	Schema, constraints, indexes and fictional-data baseline	Earlier verified baseline
SRC-08	Backend automated test suite	API, database, schema, privacy and performance assertions	109 passed; 1 performance failure
SRC-09	Frontend source and build evidence	Routes, navigation, accessibility patterns, charts and state handling	Implemented and build-validated
SRC-10	Git history	Phase, governance, implementation and release commits	Traceability evidence

1.4 Abbreviations and glossary

Term	Definition
API	Application Programming Interface
BA	Business Analyst
KPI	Key Performance Indicator
ML	Machine learning
MVP	Minimum Viable Product
NFR	Non-functional requirement
PII	Personally identifiable information
RAID	Risks, assumptions, issues and dependencies
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
V3	Third controlled quotation-acceptance dataset design

2. Executive Summary

InsureScope is an insurance market research and predictive analytics platform that consolidates fictional customer, quotation, policy, campaign and market data into a SQL-connected web application. The completed Market Research MVP gives authorised demonstration users aggregate views of product, category, region, purchase-channel and business-defined customer-segment performance.

The business problem addressed is the fragmentation of insurance research information across separate datasets and manual analyses. This fragmentation makes it difficult to compare demand, conversion, premium, product and regional performance using consistent definitions. InsureScope establishes governed KPI rules, a relational data model, read-only APIs and dashboard views so that the same totals reconcile across analytical perspectives.

The predictive-ML purpose is to define a controlled quotation-to-sale outcome prediction capability. Predictive outputs are not enabled in the released Market Research MVP. Dataset V3 generation, model fitting, challenge-label access and production insurance use remain prohibited pending separate approval. No generative-AI functionality was implemented.

The final tested implementation contains nine named frontend routes plus controlled not-found handling, eight read-only API endpoints, a fictional SQL-backed dataset, accessible loading and error states, and release evidence. Backend testing recorded 109 passing tests and one accepted performance failure. The customer-segment endpoint is functionally correct but takes approximately 25 seconds locally, against the five-second analysis target.

The Technical Business Analyst contribution connected business questions, insurance rules, data structures, API contracts, acceptance criteria, NFRs, predictive-ML controls, challenge-data confidentiality, testing evidence and release governance. Technical implementation was completed through the project using development tools and AI coding assistance; the Technical Business Analyst validated the outputs against approved requirements and evidence.

3. Business Background and Problem Statement

3.1 Current challenges

- Insurance market, quotation, policy, customer and campaign information can be held in separate operational or analytical sources.
- Manual reconciliation creates inconsistent KPI definitions and increases the risk of comparing unlike measures.
- Product, category, region and customer-segment performance can be difficult to assess through one governed view.
- Quotation conversion analysis can be distorted when region, status, time period or premium attribution rules are unclear.
- Predictive work introduces additional risks: target leakage, hidden challenge labels, weak reproducibility and unsupported business claims.

3.2 Business consequences

- Slower analysis and repeated manual reconciliation.
- Potentially inconsistent management information.
- Reduced confidence in product and regional comparisons.
- Difficulty tracing dashboard figures back to source rules and data.
- Risk of introducing predictive outputs without sufficient governance or human oversight.

3.3 Opportunity

InsureScope provides a governed analytical foundation in which business questions are translated into explicit data rules, APIs, dashboard components, acceptance criteria and test evidence. The platform also defines a future predictive capability without exposing unapproved model output.

4. Product Vision, Objectives and Success Criteria

4.1 Product vision

To provide insurance market, product, marketing and management teams with a reliable, traceable and extensible platform for understanding market performance and, when separately approved, supporting human-reviewed quotation-conversion analysis.

ID	Objective
BO-01	Create one governed view of aggregate market performance.
BO-02	Enable consistent comparison of products, categories, regions, channels and business-defined segments.
BO-03	Ensure KPI definitions reconcile across data, APIs and dashboard views.
BO-04	Define a controlled quotation-to-sale prediction capability without enabling unapproved automated decisions.
BO-05	Provide evidence suitable for release governance, audit and handover.
TO-01	Separate frontend, backend and database components.
TO-02	Expose read-only, versioned APIs with controlled errors.
ML-OBJ-01	Define a reproducible, leakage-controlled binary quotation-conversion use case.
ML-OBJ-02	Protect challenge membership and labels through custodian-controlled governance.

4.2 Success criteria

ID	Criterion	Final assessment
SC-01	All implemented routes and APIs return controlled responses.	Met
SC-02	Dashboard, category, product, regional, channel and segment totals reconcile.	Met
SC-03	Dashboard response is within three seconds under normal local use.	Met in sampled local evidence
SC-04	Market-analysis results return within five seconds.	Partially met; segment endpoint failed
SC-05	At least 100 concurrent users supported.	Not proven; 100 liveness requests only
SC-06	Prediction remains disabled until separately	Met

ID	Criterion	Final assessment
	approved.	
SC-07	Release decision and limitations are recorded.	Met
SC-08	No real insurance/customer data are used.	Met

4.3 Definition of completion

The project is complete when the approved Market Research MVP is committed, linted, built, smoke-tested, reconciled against business rules, documented with limitations, and formally approved as a portfolio and demonstration release candidate using fictional data. Public deployment and predictive-model execution are not part of the completion definition.

5. Scope

5.1 Final completed scope

- Public landing page and demonstration role selection.
- Executive dashboard.
- Category, product, regional and purchase-channel aggregate analysis.
- Stage 1 business-defined customer-segment analysis.
- Compliance and methodology disclosures.
- Read-only liveness, readiness and market-analysis APIs.
- Controlled loading, empty, error, retry and not-found states.
- Responsive navigation, accessible tables and textual chart values.
- Predictive-ML requirements, feature contracts, preprocessing controls and challenge-data governance documentation.

5.2 Out of scope or disabled

- Production authentication and authorisation.
- Quotation prediction scores, probabilities or recommendations.
- Model-generated segmentation, forecasting and next-best action.
- Official Dataset V3 generation, model fitting and challenge-label access.
- Customer-level automated insurance decisions.
- Search, filters, pagination, exports and report generation in the released UI.
- Public production deployment, SLA, disaster-recovery certification and regulated insurance processing.
- Generative AI.

5.3 Future opportunities

- Optimise customer-segment SQL and introduce appropriate caching.
- Add approved filters, search, pagination and export functions.
- Implement production identity, role-based access and audit controls.
- Add browser, accessibility, security and large-scale load automation.
- Proceed with predictive modelling only after separate dataset-execution and model-governance approvals.

6. Stakeholder Analysis

ID	Stakeholder	Type	Influence	Interest	Need	Communication
STK-01	Insurance market analyst	Internal user	High	High	Comparable aggregate market evidence	Dashboard reviews and requirement workshops
STK-02	Insurance product manager	Internal user	High	High	Product/category demand, premium and conversion insights	Prioritisation and acceptance reviews
STK-03	Marketing analyst/manager	Internal user	Medium	High	Channel, segment and campaign insight	Dashboard demos and metric validation
STK-04	Senior decision-maker	Internal sponsor/user	High	Medium	Concise, reliable	Release and decision

ID	Stakeholder	Type	Influence	Interest	Need	Communication
					executive measures	summaries
STK-05	Technical Business Analyst	Delivery/governance	High	High	Traceability, rules, contracts, test evidence	Requirements, decisions, UAT and governance
STK-06	Data/ML team member	Technical delivery	Medium	High	Feature, target, leakage and reproducibility rules	Contract reviews and evidence checks
STK-07	System administrator	Operational	Medium	Medium	Availability, security and recoverability	Operational readiness review
STK-08	Challenge-data custodian	Independent control	High	Medium	Confidential membership and protected labels	Controlled hand-off and audit evidence

6.1 RACI matrix

Activity	Technical BA	Technical delivery	Senior decision-maker	Custodian
Business requirements	A/R	C	C	I
Insurance KPI rules	A/R	C	C	I
Data/API implementation	C	A/R	C	I
UAT and release evidence	A/R	C	C	I
Predictive-ML contract	A/R	C	C	I
Challenge labels	A	I	C	R
Production deployment	C	R	A	I

7. Users and Personas

Persona	Goal	Problem	Information need
Insurance Market Analyst	Compare categories, products, regions and channels	Fragmented evidence and inconsistent measures	Rankings, conversion, premium, quotation and policy aggregates
Product Manager	Identify product demand and feature opportunities	Difficult cross-product comparison	Product/category measures and trend-ready structures
Marketing Analyst	Understand channel and segment performance	Limited joined-up customer and campaign evidence	Channel and business-defined segment summaries
Business Analyst	Trace business questions to implemented outputs	Ambiguous rules and weak audit trail	Definitions, requirements, API contracts and test evidence
Senior Decision-maker	Review concise performance information	Too much operational detail	Executive measures, ranked insights and disclosed limitations
Data/ML Team Member	Implement governed predictive work	Leakage, reproducibility and challenge confidentiality risks	Target, features, exclusions, splits, metrics and governance controls
System Administrator	Operate and monitor the platform	Prototype lacks production controls	Health, recovery, logging, access and monitoring requirements

8. As-Is and To-Be Analysis

8.1 Current-state process

Collect separate data -> Manually reconcile definitions -> Create one-off analysis -> Review inconsistent outputs -> Repeat for next question

8.2 Future-state process

Load governed fictional data -> Apply approved business rules -> Expose versioned aggregate APIs -> Present consistent dashboards -> Validate and trace evidence

Area	As-is	To-be
Data access	Separate sources and manual extracts	Governed relational model and read-only APIs
KPI definitions	May vary between analyses	Central business-rule catalogue
Regional attribution	Potentially based on current customer location	Quotation region for quotation measures; policy via quotation relationship
Error handling	Ad hoc	Controlled loading, error, retry and 404/405 behaviour
Predictive work	Risk of premature scoring	Disabled release output with explicit governance gates
Traceability	Limited	Objectives, requirements, rules, APIs, tests and commits linked

9. Functional Requirements

ID	Name	Rationale / description	Priority	Source	Acceptance criteria	Status	Test evidence
FR-001	Public landing page	Introduce the product and its controlled scope The system shall provide a public landing page with fictional-data and prediction-disablement disclosures.	Must	BA-ML-31	Given the application is opened, when the landing page loads, then product purpose and limitations are visible.	Implemented	Route smoke; asset checks
FR-002	Demonstration role selection	Support relevant user viewpoints without production identity The system shall allow selection of defined demonstration roles and keyboard operation.	Must	README; BA-ML-31	Given the role page, when a role is selected by keyboard or pointer, then the role context is accepted without authentication claims.	Implemented	Static inspection; route smoke
FR-003	Executive dashboard	Provide concise aggregate market measures The system shall display total customers, quotations, accepted quotations, policies, premium and ranked summaries where supplied by the API.	Must	BA foundation; BA-ML-31	Given the dashboard API succeeds, when the page loads, then aggregate measures are displayed and reconcile to API values.	Implemented	API and UAT reconciliation

CONFIDENTIAL - FICTIONAL INSURANCE DATA

ID	Name	Rationale / description	Priority	Source	Acceptance criteria	Status	Test evidence
FR-004	Category performance	Compare insurance categories The system shall display category-level quotations, accepted quotations, conversion and premium measures.	Must	Market research requirements	Given category data, when the page loads, then all approved categories are presented in a bounded aggregate view.	Implemented	API tests and route smoke
FR-005	Product performance	Compare insurance products The system shall display product-level quotations, conversion, premium and ranking measures.	Must	Market research requirements	Given product data, when the page loads, then product aggregates and ordering follow approved rules.	Implemented	API tests and reconciliation
FR-006	Regional performance	Identify regional opportunities The system shall display regional quotation and policy/premium performance using approved attribution rules.	Must	Regional rule decision	Given regional aggregates, when displayed, then quotation region drives quotation metrics and policy metrics follow the quotation-policy link.	Implemented	Database/API tests
FR-007	Purchase-channel performance	Compare acquisition channels The system shall display bounded aggregate channel performance.	Should	MVP scope	Given channel data, when loaded, then quotation and conversion totals reconcile to the dashboard.	Implemented	API smoke and reconciliation
FR-008	Customer-segment performance	Support Stage 1 business-defined segment analysis The system shall display aggregate metrics for the approved business-defined segments without customer identifiers.	Must	MVP scope	Given segment data, when returned, then totals reconcile and no customer identifier is exposed.	Implemented with performance limitation	Privacy/schema tests; PERF-001
FR-009	Compliance and methodology page	Disclose data, method and limitations The system shall provide methodology, fictional-data and disabled-prediction disclosures.	Must	BA-ML-31	Given the compliance route, when opened, then scope and limitations are clear.	Implemented	Route smoke; static inspection
FR-010	Loading state	Prevent ambiguous waiting behaviour The system shall expose accessible loading status while bounded API requests are pending.	Must	UX requirements	Given a request is pending, then a status region communicates loading.	Implemented	Source inspection
FR-011	Error state	Handle unavailable services safely The system shall show a controlled error without stack trace or confidential detail.	Must	NFR/security	Given the backend is unavailable, when a request fails, then the user sees a controlled message and retry option.	Implemented	Backend-stop UAT; 404/405 checks
FR-012	Empty state	Explain valid no-data responses The system shall distinguish a valid empty result from a failure.	Should	UX requirements	Given an empty collection, then an explicit empty state is displayed.	Implemented	Source inspection
FR-013	User-controlled retry	Avoid uncontrolled repeated requests The system shall permit manual retry and shall not automatically loop failed requests.	Must	Traffic/reliability requirements	Given a failed request, when Retry is selected, then one new request is issued.	Implemented	Source inspection and recovery UAT

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ID	Name	Rationale / description	Priority	Source	Acceptance criteria	Status	Test evidence
FR-014	Controlled not-found page	Support invalid navigation The system shall provide a wildcard not-found route.	Should	Frontend requirements	Given an unsupported frontend route, then a controlled not-found page is shown.	Implemented	Route smoke
FR-015	Read-only market APIs	Protect prototype data The system shall expose implemented market endpoints as GET only.	Must	API contract	Given a supported endpoint, GET returns 200; unsupported POST returns 405.	Implemented	API smoke
FR-016	Liveness check	Support operational health checks The system shall provide a process liveness endpoint.	Must	Operational requirements	Given the process is running, GET /health/live returns 200.	Implemented	Smoke and 100-request check
FR-017	Readiness check	Confirm database availability The system shall provide a database readiness endpoint.	Must	Operational requirements	Given the database connection is available, GET /health/ready returns 200.	Implemented	Smoke and startup check
FR-018	Consistent quotation totals	Prevent conflicting management information All applicable aggregate APIs shall return the same total quotation count for the same unfiltered dataset.	Must	KPI rules	Given the same dataset, totals equal 10,000 across six views.	Implemented	Cross-API reconciliation
FR-019	Consistent accepted totals	Prevent conflicting conversion measures All applicable aggregate APIs shall return the same accepted quotation count.	Must	KPI rules	Given the same dataset, accepted totals equal 5,077 across six views.	Implemented	Cross-API reconciliation
FR-020	Fictional-data notice	Prevent misrepresentation The system shall display a shared fictional-data notice throughout the application shell and public pages.	Must	Privacy/governance	Given any page is displayed, then the fictional-data notice is visible.	Implemented	Static inspection and release review
FR-021	No customer identifiers in segment API	Protect privacy The customer-segment endpoint shall expose aggregates only.	Must	Privacy requirement	Given segment data is returned, then no customer identifier or reference is present.	Implemented	Schema/privacy tests
FR-022	Prediction disabled	Enforce approved release boundary The application shall not expose prediction routes, endpoints, scores, probabilities or automated decisions.	Must	BA-ML-31	Given the released application, when routes and endpoints are inspected, then no predictive function is registered or displayed.	Implemented	Source scan; route/API inventory
FR-023	Quotation-conversion target contract	Define future binary outcome The ML design shall map Accepted to 1 and Rejected/Expired to 0; Pending shall be excluded from model-ready labels.	Must	BA-ML-27	Given an eligible quotation, when labels are constructed, then the approved mapping is applied and Pending has no target.	Approved requirement; execution blocked	Contract and non-generation tests
FR-024	Leakage control	Protect model validity Post-outcome, policy and generated-probability fields shall be excluded from features.	Must	BA-ML-27/29	Given a feature artifact, when validated, then denylisted fields are absent.	Approved; temporary implementation evidence	Schema validation tests

CONFIDENTIAL - FICTIONAL INSURANCE DATA

ID	Name	Rationale / description	Priority	Source	Acceptance criteria	Status	Test evidence
FR-025	Challenge confidentiality	Prevent inference of challenge membership Challenge Pending membership shall remain confidential and be derived using a custodian-held secret value outside Git.	Must	BA-ML-28	Given public artifacts, when inspected, then secret derivation values and challenge membership are not exposed.	Approved governance	Commitment/hash controls
FR-026	Protected challenge labels	Prevent unauthorised model evaluation Challenge labels shall be generated and held only through the controlled custodian workflow.	Must	BA-ML-27/28	Given normal project access, then protected challenge labels cannot be read or generated.	Approved; not executed	Governance review
FR-027	Reproducible dataset identity	Support audit and repeatability Dataset versions, seeds, namespaces, counts and fingerprints shall be frozen and version-controlled without exposing secrets.	Must	BA-ML-27	Given the same authorised configuration and secret control, deterministic membership evidence can be reproduced.	Approved; final generation blocked	Contract tests
FR-028	Separate development and challenge populations	Prevent contamination Development and challenge identities, seeds and namespaces shall not overlap.	Must	BA-ML-27	Given both packages, then customer, quotation and reference memberships do not overlap.	Approved	Overlap validation
FR-029	Feature schema contract	Create model-ready, governed inputs The ML feature set shall use approved pre-decision fields and documented types, ranges and null rules.	Must	BA-ML-29	Given temporary rows, when validated, then every feature matches the frozen schema.	Implemented for temporary controlled work	Feature-contract tests
FR-030	Preprocessing contract	Provide deterministic transformations Preprocessing shall define type handling, encoding, scaling where required, missing-value rules and stable output order.	Must	BA-ML-30	Given equivalent input and frozen configuration, then preprocessing output is reproducible.	Implemented for temporary controlled work	Preprocessing tests
FR-031	Explainability requirements	Support human interpretation Any future model shall provide global and local explanations using business-readable feature descriptions.	Must	Predictive-ML requirements	Given a prediction is later authorised, then users can see the principal contributing factors without causal claims.	Planned; model not approved	Evidence not available
FR-032	Human oversight	Prevent automated insurance decisions Predictions shall support review and shall not directly determine acceptance, pricing or coverage.	Must	Governance requirements	Given an authorised score, then a human decision owner remains accountable.	Approved requirement; not implemented	Evidence not available
FR-033	Search and filtering	Support focused analysis The system should support approved filters and search parameters.	Could	Earlier scope/NFR	Given filters are implemented, then API and UI results reflect validated inputs.	Not implemented	Not tested
FR-034	Pagination	Protect performance for large results Large customer-level or reference lists	Should	NFR	Given a large result set, then only the requested page is	Not applicable to bounded aggregate MVP	Not tested

ID	Name	Rationale / description	Priority	Source	Acceptance criteria	Status	Test evidence
		shall be paginated.			returned.		
FR-035	Reporting/export	Support controlled distribution The system may generate or export approved reports in a background process.	Could	NFR	Given a large report request, then it does not block the dashboard.	Not implemented	Not tested
FR-036	Audit and monitoring	Support operational assurance Production implementation shall record service, API, ML usage and error evidence without confidential customer content.	Must for production	NFR	Given an operational event, then appropriate metadata is logged securely.	Recommended; not implemented	Not tested

10. Non-Functional Requirements

ID	Category	Requirement	Assessment
NFR-001	Performance	Dashboard pages shall load within 3 seconds under normal usage.	Validated locally for preview and dashboard API; browser page timing not fully measured.
NFR-002	Performance	Search and market-analysis results shall appear within 5 seconds, excluding external AI delays.	Partially met; customer-segment API approximately 25 seconds.
NFR-003	Capacity	Prototype shall support at least 100 concurrent users.	Not proven; 100 concurrent liveness requests passed.
NFR-004	Scalability	Production design shall allow future scaling to 1,000 concurrent users.	Architecture recommendation; not load-certified.
NFR-005	Rate limiting	Repeated requests shall be limited by user and function.	Not implemented.
NFR-006	Queuing	Expensive ML/report requests shall use a queue during high traffic.	Not implemented; no such request enabled.
NFR-007	Caching	Frequently requested market data shall be cached.	Not implemented as production cache.
NFR-008	Pagination	Large datasets shall use pagination.	Not implemented; current APIs return bounded aggregate views.
NFR-009	Database indexing	Frequently searched insurer, product, country, segment and date fields shall be indexed.	Earlier schema baseline records indexes; final index audit not included in release evidence.
NFR-010	Efficient retrieval	The application shall avoid loading complete datasets when only aggregates or limited rows are required.	Implemented through aggregate SQL APIs.
NFR-011	Controlled retries	External/API failures shall use controlled retries.	User-controlled retry implemented; no automatic retry loop.
NFR-012	Background reporting	Large reports shall be generated in the background.	Not implemented.
NFR-013	Availability	Production target shall be at least 99.5% monthly availability.	Target only; no production hosting/SLA.
NFR-014	Failure isolation	Service failure shall not crash the complete application.	Controlled frontend unavailable states observed.
NFR-015	Backup	Database backups shall run at least daily.	Not configured or validated.

ID	Category	Requirement	Assessment
NFR-016	Recovery	A recovery procedure shall be documented and tested.	Local restart recovery documented and tested; production recovery not certified.
NFR-017	Monitoring	Response time, CPU, memory, database performance and API failures shall be monitored.	Not implemented as production observability.
NFR-018	CPU alert	Alert when CPU exceeds 70% for a sustained period.	Recommendation; not implemented.
NFR-019	Secure logging	Errors shall be recorded without confidential customer information.	Controlled errors and source scan support prototype handling; production retention not assessed.
NFR-020	ML usage/cost monitoring	ML API or compute usage and cost shall be monitored separately where applicable.	Not applicable to disabled prediction; production recommendation.
NFR-021	Load test - 100	Test 100 simultaneous users and record latency/failure/recovery.	Only 100 simultaneous liveness requests performed.
NFR-022	Load test - 500	Test 500 simultaneous users.	Not performed.
NFR-023	Dependency failure	Test slow/unavailable external APIs and database connection limits.	Backend unavailable/restart tested; broader failure scenarios not performed.
NFR-024	Security	Production use requires authentication, authorisation, secrets, vulnerability, privacy and penetration testing.	Not implemented; production blocker.

10.1 Implemented and validated

- Frontend production build and API smoke.
- Controlled loading, error, empty and retry behaviour.
- Read-only bounded aggregate API design.
- Dashboard and most API response times within target in local samples.
- Local restart recovery and 100 simultaneous liveness-request smoke.

10.2 Production recommendations not implemented

- Identity and role-based access.
- Global rate limiting, queueing and infrastructure caching.
- Observability, 70% CPU alerts and secure retention policies.
- Daily backups, disaster recovery and 99.5% availability controls.
- 500-user load testing and database-connection saturation testing.

11. Business Rules and KPI Definitions

ID	Rule	Definition	Example
BR-001	Quotation status	Allowed status values are Accepted, Rejected, Expired and Pending within the relevant governed dataset.	A quotation marked Accepted is included in accepted counts.
BR-002	Accepted quotation	A quotation is accepted only where quotation status equals Accepted.	Accepted quotations = 5,077 in the released fictional market dataset.
BR-003	Conversion rate	Accepted quotations / total eligible quotations x 100. For released dashboard aggregates, use the API-defined total quotation denominator unless a filtered rule is approved.	$5,077 / 10,000 \times 100 = 50.77\%$.
BR-004	Written premium	Sum of written premium for policies created through accepted quotations, following the policy-to-quotation relationship.	A policy premium contributes to the product/category/region of its originating quotation.

ID	Rule	Definition	Example
BR-005	Average premium	Written premium / applicable policy count. Return zero or null according to the approved API schema when count is zero.	£500,000 / 1,000 policies = £500.
BR-006	Policy count	Count distinct policy records associated with accepted quotations.	Released baseline policy count: 5,077.
BR-007	Quotation regional attribution	Quotation measures use the quotation region, not the customer current region.	A customer who moved after quoting does not change historical quotation attribution.
BR-008	Policy/premium regional attribution	Policy and premium measures follow policy -> quotation -> quotation region.	The originating quotation region is retained.
BR-009	Product/category attribution	A quotation is attributed through its selected product and the product category relationship.	Product totals roll up to category totals.
BR-010	Ranking	Sort by the approved primary measure descending, with a stable secondary key for ties.	Two equal conversion rates use the defined secondary order.
BR-011	Zero activity	Approved reference items may appear with zero measures when required by the API contract; division-by-zero is controlled.	A category with no quotations shows zero rather than an error.
BR-012	Date filtering	Date boundaries are inclusive unless the API contract explicitly states otherwise.	1 January through 31 March includes both dates.
BR-013	Customer segments	Released segments are Stage 1 business-defined groupings, not model-generated segments or next-best actions.	Segment labels describe research groupings only.
BR-014	Prediction label	Accepted = 1; Rejected/Expired = 0; Pending excluded from model-ready population and labels.	Pending receives no binary target.

12. Data Requirements

12.1 Conceptual model

Customer -> Quotation -> Product -> Category

Quotation -> Policy -> Premium

Customer / Quotation -> Region / Segment / Channel -> Aggregate APIs

12.2 Main entities and relationships

Entity	Purpose	Relationship
Customer	Fictional customer profile and business-defined segment	One customer may create many quotations.
Quotation	Insurance offer, date, region, product, channel, premium and status	Belongs to one customer and one product; may create one policy.
Policy	Written policy created from an accepted quotation	References originating quotation.
Product	Insurance product offered for quotation	Belongs to one category and may have features/add-ons.
Category	Insurance category reference	Groups products.
Region	Quotation-market region reference	Quotation region controls quotation attribution.

Entity	Purpose	Relationship
Customer Segment	Stage 1 business-defined grouping	Used only for aggregate research views.
Purchase Channel	Acquisition channel reference	Used for aggregate channel analysis.
Competitor	Fictional competitor reference	Earlier baseline; not exposed as a final MVP route.
Campaign	Fictional campaign reference	Monitoring/research context; no target effect in V3 contract.

12.3 Verified and earlier baseline figures

Measure	Figure	Evidence status
Customers	5,000	Verified in final cross-API evidence
Quotations	10,000	Verified in final cross-API evidence
Accepted quotations	5,077	Verified in final cross-API evidence
Policies	5,077	Earlier baseline and consistent with accepted total; final release record does not separately restate policy count
Insurance categories	8	Current README and earlier dataset baseline
Customer segments	8	Earlier approved baseline; final release confirms Stage 1 segments but not final reference-count audit
Regions	12	Earlier approved baseline; final release does not repeat count
Products	33	Earlier approved baseline; final release does not repeat count
Product features	66	Earlier approved baseline; final release does not repeat count
Add-ons	40	Earlier approved baseline; final release does not repeat count
Fictional competitors	5	Earlier approved baseline; not exposed in final route inventory
Tables / FKs / checks / indexes	29 / 45 / 33 / 28	Earlier database baseline; final database re-count evidence not available in BA-ML-31

12.4 Data dictionary - core fields

Field	Entity	Definition	Validation
customer_id	Customer	Unique fictional customer key	Not null; unique; not exposed by segment API
quotation_id	Quotation	Unique fictional quotation key	Not null; unique
quotation_date	Quotation	Date quotation was created	Valid date; used for period attribution

Field	Entity	Definition	Validation
quotation_status	Quotation	Accepted, Rejected, Expired or Pending	Approved values only
quotation_region_id	Quotation	Region at time of quotation	Mandatory for regional quotation attribution
product_id	Quotation/Product	Selected insurance product	Valid foreign key
category_id	Product/Category	Insurance category of product	Derived through product relationship
quoted_premium	Quotation	Premium offered	Non-negative; currency precision controlled
purchase_channel_id	Quotation	Acquisition channel	Valid reference value
policy_id	Policy	Unique policy key	Created only through accepted quotation in released data
written_premium	Policy	Premium written on policy	Non-negative; aggregated through originating quotation
segment_id	Customer/Segment	Business-defined segment key	Used only in aggregate output
target	ML label	Binary conversion outcome	1 Accepted; 0 Rejected/Expired; absent for Pending

12.5 Data quality, privacy and lineage

- Primary and foreign keys must preserve referential integrity.
- Status, category, product, region, channel and segment values must conform to approved reference data.
- Duplicate business keys are rejected or quarantined before loading.
- Missing required relationship keys fail validation; optional missing values follow documented rules.
- All data are fictional; no real customer or insurer records are permitted.
- Customer identifiers must not be exposed by aggregate analytical APIs.
- Lineage is maintained from source fictional tables through SQL aggregation, API response and dashboard component.
- ML feature-source mapping must identify source field, business meaning, transformation, leakage classification and availability time.

13. Technical Architecture

13.1 Implemented architecture

Layer	Technology	Responsibility
Frontend	React, Vite, TypeScript/JavaScript source, Recharts	Routes, pages, tables, charts, responsive and accessibility behaviour
Backend	Python, FastAPI	Read-only health and aggregate market APIs
Database	SQL Server	Relational fictional insurance data and aggregate queries
Data/ML	Pandas, scikit-learn and controlled Python modules	Feature and preprocessing contracts; released prediction disabled
Testing	pytest; ESLint; Vite build	Backend automated tests, lint, build and release smoke
Version control	Git and GitHub	Requirements, implementation and approval history
Power BI	Project technology area	Evidence not available that Power BI formed part of the final released web MVP

13.2 System context

Business user -> React web application -> FastAPI read-only APIs -> SQL Server fictional dataset

13.3 Predictive workflow boundary

Approved feature contract -> Controlled temporary generation -> Approved preprocessing -> Separate future model approval -> Human-reviewed use

The released application stops before model training, prediction and score presentation. No ML module is imported into the market-research dashboard path.

13.4 Deployment approach

The final release evidence confirms a local prototype and production-preview build. At approval time there was no configured public deployment target, authentication workflow or rollback automation. GitHub now preserves the committed repository, but repository hosting is not equivalent to application deployment.

14. API Requirements and Catalogue

Method	Route	Purpose	Parameters	Response	Statuses	Rules	Evidence
GET	/api/v1/health/live	Process liveness	None	Health status	200; controlled server errors	Operational check	Smoke; 100 concurrent requests
GET	/api/v1/health/ready	Database readiness	None	Readiness status	200 when database available	Database connection	Smoke and startup
GET	/api/v1/dashboard/summary	Executive aggregate measures	None	Customers, quotations, accepted totals and dashboard summaries	200; controlled error	BR-002 to BR-006	Automated tests; 10-sample timing
GET	/api/v1/market/category-performance	Category aggregates	None	Bounded category measures	200; controlled error	BR-003, BR-009	Automated tests; reconciliation
GET	/api/v1/market/product-performance	Product aggregates	None	Bounded product measures and ranking	200; controlled error	BR-003, BR-009, BR-010	Automated tests; reconciliation
GET	/api/v1/market/regional-performance	Regional aggregates	None	Quotation, policy and premium regional measures	200; controlled error	BR-007, BR-008	Automated tests; reconciliation
GET	/api/v1/market/purchase-channel-performance	Channel aggregates	None	Quotation and conversion channel measures	200; controlled error	Approved channel attribution	Automated tests; reconciliation
GET	/api/v1/market/customer-segment-performance	Stage 1 segment aggregates	None	Aggregate segment counts and measures; no customer identifiers	200; controlled error	BR-013; privacy rule	Schema/privacy tests; PERF-001

The current APIs accept no search, filter or pagination query parameters. Unsupported routes return 404 and unsupported POST requests return 405 without stack traces or confidential detail.

15. Predictive-ML Requirements

Area	Requirement
Business question	Which eligible quotations are more likely to convert to a sale, to support prioritised human review?
Target variable	Binary quotation acceptance outcome.
Converted	Quotation status Accepted = 1.

Area	Requirement
Not converted	Rejected or Expired = 0.
Excluded from labels	Pending; it remains in the complete population only.
Prediction population	Eligible pre-decision quotations within the approved fictional development/challenge populations.
Input features	Approved pre-decision customer, quotation, product, region, channel, affordability and contextual fields defined by the feature contract.
Excluded features	Outcome, rejection reason, policy-created fields, written premium, acceptance date, generated probability and other post-decision/leakage fields.
Development/challenge design	Separate seeds, namespaces and non-overlapping memberships; challenge labels protected.
Class balance	Must be measured from authorised model-ready data; final model evidence not available.
Candidate algorithms	Logistic regression baseline and suitable tree-based classifiers may be evaluated after model authorisation.
Baseline model	Evidence not available for an approved final model.
Model selection	Compare accuracy, precision, recall, F1, ROC-AUC, calibration, stability, fairness and business usefulness.
Threshold selection	Must be based on business costs, review capacity and false-positive/false-negative trade-offs; not fixed by accuracy alone.
Explainability	Global and local explanations with business-readable feature names; no causal claims.
Fairness	Evaluate performance across relevant approved groups; investigate material disparities before use.
Reproducibility	Frozen version, configuration, seeds, code hashes, manifests and environment evidence.
Monitoring	Track input drift, output distribution, performance when labels mature, fairness and operational failures.
Retraining	Triggered by approved drift/performance thresholds, material data or product change, or governance review.
Human oversight	No automated acceptance, pricing, coverage or adverse customer decision.
Current status	Prediction disabled; no approved final model results.

15.1 Model metrics

Accuracy, precision, recall, F1-score and ROC-AUC are required evaluation measures. Final numerical model results are not included because no final model was approved or exposed. Any future result must state dataset version, split, threshold, confidence interval or repeated-run method where applicable, and the authorised evaluator.

16. Challenge-Data Governance

16.1 Confidentiality problem

Challenge Pending membership must remain confidential. A public deterministic SHA-256 selection rule is insufficient when every required input is public, because an observer can reproduce the selection. Removing Pending ordinals from a public document also does not protect membership when the deterministic rule and complete identifier space remain available.

16.2 Approved Option A

- Use a custodian-held secret derivation value stored outside Git and outside normal developer access.
- Publish only approved commitment hashes, counts and non-sensitive reproducibility evidence.
- Do not publish the secret value, Pending identifiers, protected challenge labels or derivation material.
- Amend BA-ML-27 so public deterministic evidence does not claim confidentiality without the secret control.
- Require independent custodian reproduction and audit evidence before protected label generation.

16.3 Roles and controls

Role	Responsibility
Technical Business Analyst	Owns requirements, governance decisions, approval gates and traceability; does not access protected challenge labels without separate authority.
Technical delivery	Implements approved public and controlled components; cannot embed or log secret values.
Challenge-data custodian	Holds secret derivation value, reproduces eligible membership, generates protected labels and preserves audit evidence.
Reviewer	Verifies public commitments, counts, fingerprints and control operation without receiving unnecessary secret material.

Security limitation: secret-based deterministic selection improves confidentiality but still requires secure storage, access control, rotation/recovery procedures and independent operational assurance. Those production controls were not certified in the released prototype.

17. Dashboard and UX Requirements

Area	Requirement / evidence
Information architecture	Landing, roles, dashboard, categories, products, regions, channels, segments, compliance and controlled not-found page.
Navigation	Responsive labelled navigation with mobile close behaviour and skip link.
Tables	Captions, row/column headers and horizontal scrolling for wide content.
Charts	Recharts visualisations accompanied by textual values.
Ranking	Stable, business-rule-aligned ordering.
Prediction displays	Not present; explicit disabled status.
Loading	Accessible status region.
Empty	Explicit no-data state.
Error	Controlled alert with user Retry.
Accessibility	Semantic headings, focus-visible styles, keyboard role selection and reduced-motion support.
Responsive behaviour	Desktop, tablet and mobile media-query ranges.

Area	Requirement / evidence
Performance	Dashboard target 3 seconds; market-analysis target 5 seconds, with disclosed segment exception.
Unverified UX evidence	Route-specific titles, visual viewport inspection, measured contrast, screen-reader testing and browser-console review.

18. Assumptions, Dependencies and Constraints

Type	Statement
Assumption	All data remain fictional and suitable only for controlled demonstration.
Assumption	SQL Server is available for local API operation.
Dependency	Frontend depends on read-only FastAPI endpoints and agreed schemas.
Dependency	Backend depends on SQL Server schema, data and connectivity.
Dependency	Predictive work depends on separate Dataset V3 execution and model approvals.
Constraint	No production authentication, authorisation, observability, rate limiting, backup or SLA.
Constraint	No real insurer/customer data, regulated decisioning or production traffic.
Constraint	Customer-segment endpoint materially exceeds the five-second target.
Constraint	Search, filter, pagination, export and report generation are absent.
Constraint	Challenge secrets and labels must remain outside Git and normal project access.
Deployment assumption	GitHub stores committed source and history; a public live application is not yet configured.

19. RAID Log

ID	Risk	Impact	Probability	Mitigation	Status	Evidence
R-01	Segment API latency affects demonstration experience	High	High	Optimise query, indexes and caching; disclose current latency	Open	PERF-001
R-02	Public repository may be mistaken for production readiness	High	Medium	Prominent fictional-data and limitation statements	Controlled	README; BA-ML-31
R-03	Challenge membership disclosure undermines evaluation integrity	High	Medium	Custodian-held secret derivation value and public commitments only	Controlled by approved design	BA-ML-28
R-04	Predictive output used without approval	High	Low	No route, endpoint, score or model integration	Controlled	BA-ML-31

CONFIDENTIAL - FICTIONAL INSURANCE DATA

ID	Risk	Impact	Probability	Mitigation	Status	Evidence
R-05	Lack of production identity/security	High	High	Block production use until controls and testing exist	Open production blocker	BA-ML-31

ID	Assumption	Owner	Impact	Status	Evidence
A-01	Fictional data sufficiently demonstrates approved analytical behaviour	Technical BA	Medium	Accepted for approved use	README/BA-ML-31
A-02	Local environment reflects functional integration but not production scale	Technical BA	High	Explicit limitation	BA-ML-31
A-03	Earlier database counts remain unchanged where no later contradictory evidence exists	Technical BA	Medium	Mark as earlier baseline, not final re-count	Source register

ID	Issue	Owner	Impact	Resolution / action	Status	Evidence
I-01	Customer-segment API approximately 25 seconds	Technical delivery	High	Accepted prototype limitation; post-release optimisation	Open	PERF-001
I-02	No browser/front-end automated test script	Technical delivery	Medium	Introduce Vitest/RTL/browser checks in next phase	Open	BA-ML-31
I-03	No public deployment workflow	Repository owner	Medium	Define zero-cost or controlled hosting architecture separately	Open	BA-ML-31

ID	Dependency	Affects	Status
D-01	SQL Server availability	Backend and analytical APIs	Active for local prototype
D-02	Approved API schemas	Frontend pages	Implemented
D-03	Custodian secret and secure process	Challenge confidentiality and labels	Not executed
D-04	Separate approval	Dataset V3 execution and model fitting	Blocked
D-05	Production infrastructure controls	Public production use	Not available

20. Decision Log

ID	Date	Decision	Rationale	Alternative/trigger	Approver	Impact
DEC-01	xx Feb 20xx	Product name	Use InsureScope as approved name; MarketLens retained only as an earlier working name.	Current product identity	Technical Business Analyst	Consistent professional documentation
DEC-02	xx Feb 20xx	Release scope	Release aggregate market-research MVP; prediction disabled.	Avoid unsupported claims and model exposure	Technical Business Analyst	Clear product boundary
DEC-03	xx Feb 20xx	Technology	React/Vite frontend, FastAPI/Python	Separation and compatibility with	Technical Business Analyst / delivery	Three-layer architecture

ID	Date	Decision	Rationale	Alternative/trigger	Approver	Impact
			backend and SQL Server database.	approved local stack	evidence	
DEC-04	xx Feb 20xx	Regional attribution	Use quotation region for quotation metrics; policy/premium through originating quotation.	Preserve historical transaction context	Technical Business Analyst	Consistent regional KPIs
DEC-05	xx Feb 20xx	API contract	Read-only versioned GET endpoints; no unsupported filters.	Match bounded aggregate MVP	Technical Business Analyst	Controlled frontend/API alignment
DEC-06	xx Feb 20xx	Generative AI	Do not include generative AI in implemented scope.	No implementation evidence	Technical Business Analyst	No GenAI claim
DEC-07	xx Feb 20xx	Predictive use case	Quotation-to-sale binary outcome remains the core future ML use case.	Business relevance and controlled scope	Technical Business Analyst	Requirements and governance only
DEC-08	xx Feb 20xx	Challenge confidentiality	Use custodian-held secret derivation value outside Git.	Public deterministic hashes alone do not protect membership	Technical Business Analyst	Approved Option A
DEC-09	xx Feb 20xx	Release performance	Accept PERF-001 as non-blocking prototype limitation.	Late rewrite would increase correctness/governance risk	Technical Business Analyst	Release approved with disclosure
DEC-10	xx Feb 20xx	Approval	Approve portfolio and demonstration release candidate using fictional data.	Required release closure	Technical Business Analyst	Commit ea26f2e

21. Requirements Traceability Matrix

Objective	Business req.	Functional req.	Rule	API/component	Acceptance	Test	Status
BO-01	BRQ-01	FR-003	BR-002 to BR-006	Dashboard / summary API	AC-FR-003	TC-API-03	Passed
BO-02	BRQ-02	FR-004	BR-003/009	Category page/API	AC-FR-004	TC-API-04	Passed
BO-02	BRQ-03	FR-005	BR-003/009/010	Product page/API	AC-FR-005	TC-API-05	Passed
BO-02	BRQ-04	FR-006	BR-007/008	Region page/API	AC-FR-006	TC-API-06	Passed
BO-02	BRQ-05	FR-007	Channel attribution	Channel page/API	AC-FR-007	TC-API-07	Passed
BO-02	BRQ-06	FR-008	BR-013	Segment page/API	AC-FR-008	TC-PERF-01	Functional pass; performance fail
BO-03	BRQ-07	FR-018/019	BR-002/003	All aggregate APIs	AC-RECON	TC-REC-01	Passed
BO-04	BRQ-08	FR-022	Release boundary	Router/API registration	AC-NOPRED	TC-SEC-01	Passed
ML-OBJ-01	BRQ-09	FR-023/024	BR-014	ML contracts	AC-ML-LABEL	TC-ML-01	Approved non-generation evidence
ML-OBJ-02	BRQ-10	FR-025/026	Challenge confidentiality	Custodian workflow	AC-CHL	TC-GOV-01	Approved design; not executed
TO-01	BRQ-11	FR-015 to 017	Health/API rules	FastAPI	AC-HEALTH	TC-API-01/02	Passed
BO-05	BRQ-12	FR-020/036	Disclosure/audit	Shared notice / release docs	AC-DISC	TC-REL-01	Passed / production monitoring open

22. Testing and UAT

22.1 Strategy

- Validate business rules and schema through automated backend tests.
- Exercise each health and market endpoint using the real local fictional SQL database.
- Reconcile totals across dashboard and analytical perspectives.
- Validate privacy by asserting aggregate schemas and absence of customer identifiers.
- Run lint and production build for frontend regression evidence.
- Use route, asset, service-failure and restart smoke for release UAT.
- Treat missing browser automation, security certification and large-scale load evidence as explicit gaps.

Test ID	Test / evidence	Result
TC-API-01	Liveness endpoint returns 200	Passed
TC-API-02	Readiness endpoint returns 200 with database	Passed
TC-API-03	Dashboard summary schema and totals	Passed
TC-API-04	Category performance schema/totals/order	Passed
TC-API-05	Product performance schema/totals/order	Passed
TC-API-06	Regional performance and attribution	Passed
TC-API-07	Purchase-channel performance	Passed
TC-API-08	Customer-segment aggregate schema/privacy	Passed
TC-API-09	Unsupported route returns 404	Passed
TC-API-10	Unsupported POST returns 405	Passed
TC-REC-01	10,000 quotations reconcile across six views	Passed
TC-REC-02	5,077 accepted quotations reconcile across six views	Passed
TC-REC-03	5,000 customers reconcile dashboard to segment sum	Passed
TC-PERF-01	Customer-segment response under five seconds	Failed; 25.30-25.43 seconds
TC-PERF-02	Dashboard 10-sample response timing	Passed; max 116.12 ms
TC-LOAD-01	100 simultaneous liveness requests	Passed; 100/100 in 561.59 ms
TC-RECOV-01	Backend restart recovery	Passed; liveness restored in 2.165 seconds
TC-FE-01	All named routes and wildcard return application shell	Passed by HTTP smoke
TC-FE-02	JavaScript, CSS and favicon return 200	Passed
TC-FE-03	ESLint	Passed
TC-FE-04	Production build	Passed; 99 modules; Vite 525 ms
TC-SEC-01	No prediction route/endpoint/score/model call	Passed
TC-SEC-02	Tracked source/bundle secret and PII scan	Passed within stated scope
TC-UX-01	Loading/empty/error/retry states present	Passed by source and service-failure review
TC-A11Y-01	Semantic static accessibility checks	Passed static review; browser certification absent

Test ID	Test / evidence	Result
TC-DEP-01	Python dependency consistency	Passed: pip check
TC-DEP-02	Production npm high-severity audit	Passed: 0 vulnerabilities

Automated backend result: 109 passed, 1 failed and 1 warning in 488.15 seconds. The single failure was PERF-001. No configured frontend test script or browser automation existed, so those passes are not claimed.

23. Delivery History

Phase	Deliverables	Blocker / issue	Resolution / evidence
Discovery	Defined insurance market-research problem, target users and product questions.	Scope ambiguity	Established eight insurance categories and governed objectives.
BA foundation	Created requirements, processes, business rules, NFRs and traceability structure.	Need to separate BA ownership from technical implementation	Documented roles, acceptance and evidence boundaries.
Environment setup	Verified Python, npm, virtual environment and repository controls.	SQL Server trust/certificate issue	Recorded environment and connection constraints.
Database design	Created relational schema, constraints, indexes and fictional-data requirements.	Complex KPI attribution	Defined quotation-region and policy-through-quotation rules.
Fictional data	Generated and loaded customer, quotation, policy and reference data.	Need consistent totals	Reconciled 10,000 quotations and 5,077 accepted quotations.
Backend APIs	Implemented health and aggregate market endpoints.	Schema and error consistency	Used read-only versioned contracts and automated tests.
Dashboard development	Implemented executive and analysis pages, states and disclosures.	Prediction wording exceeded implementation	Corrected scope and fictional-data notice.
Predictive-ML work	Defined V1/V2/V3 requirements, feature and preprocessing contracts.	No approved final model	Kept model execution and prediction disabled.
Challenge governance	Identified reproducibility/confidentiality conflict.	Public deterministic selection could expose membership	Approved custodian-secret Option A.
Testing and completion	Ran backend, API, build, lint, performance, privacy and recovery checks.	Segment endpoint missed target	Accepted disclosed limitation and approved release candidate.

24. Business Value and Outcomes

- Translated insurance market questions into implemented, testable analytical views.
- Created traceability between KPI rules, database relationships, APIs and dashboard components.
- Reduced ambiguity by defining accepted quotation, conversion, premium, policy and regional attribution rules.
- Prevented incorrect regional reporting by separating quotation region from a customer current region.
- Defined controlled acceptance criteria and used test evidence to challenge unsupported completion claims.
- Established a governed future quotation-conversion use case without exposing unapproved scores.
- Protected challenge evaluation integrity by identifying the weakness in public deterministic membership selection.
- Created a separable frontend, backend and database foundation for future scaling work.
- Accelerated delivery through early requirements, prototype validation and iterative evidence review using development tools and AI coding assistants.

No financial saving, conversion uplift or productivity percentage is claimed because outcome evidence is not available.

25. Limitations and Production Boundary

The portfolio prototype demonstrates product functionality and Technical Business Analyst capability. It is not intended to support production-level insurance data or large-scale public traffic without additional security, infrastructure, performance, legal, compliance and operational testing.

- No production authentication, authorisation, rate limiting, monitoring, backups or SLA.
- No real customer or insurer data.
- No public deployment workflow at final approval.
- No production security, privacy, regulatory or penetration certification.
- No final model, score, probability, forecasting or automated decision capability.
- Customer-segment API materially misses the market-analysis response target.
- No full browser, screen-reader, contrast or 500-user load certification.

26. Lessons Learned and Recommendations

Area	Lesson
Requirements	Separate current release scope from future capability; wording must match tested implementation.
Business rules	Define attribution and denominator rules before API and dashboard development.
Data governance	Reproducibility does not automatically provide confidentiality; public deterministic inputs can expose membership.
API contracts	Do not expose filters or controls until parameter validation and response behaviour are implemented.
ML governance	Feature and preprocessing contracts can be approved before model work, while generation and model authority remain separate gates.
Testing	A large passing count does not cancel a material performance failure; record the failure and business decision explicitly.
Scalability	A 100-request liveness smoke is not evidence of 100-user capacity; realistic journeys and database pressure must be tested.
Release	Correct narrow scope defects rather than rewriting architecture late in release preparation.

26.1 Production recommendations

- Optimise customer-segment SQL; review execution plans and indexes; introduce cache with controlled invalidation.
- Implement identity, role-based access, secrets management, rate limits and secure audit logging.
- Add API parameter contracts for approved filters, search and pagination.
- Add Vitest/React Testing Library and browser-based accessibility/regression automation.
- Establish deployment, rollback, backup, recovery, observability and 99.5% service controls.
- Complete 100- and 500-user journey-based load tests, external failure scenarios and database connection-limit tests.
- Keep Dataset V3, challenge labels and model fitting behind separate written approvals.

27. Technical Business Analyst Contribution Summary

The Technical Business Analyst owned the translation of the insurance business problem into a governed delivery structure. This included requirements elicitation and analysis, user and stakeholder needs, current/future process analysis, business rules, KPI definitions, data and API requirements, acceptance criteria, NFRs, RAID management, decision records, traceability, predictive-ML requirements, explainability and human-oversight requirements, challenge-data governance, UAT evidence review and release acceptance.

The Technical Business Analyst connected business intent to technical evidence by checking that database relationships supported agreed attribution rules, API schemas matched dashboard needs, aggregate totals reconciled, privacy controls prevented customer-level exposure, and implemented wording did not imply unavailable prediction capability.

Technical implementation was completed through the project using development tools and AI coding assistants. The Technical Business Analyst directed, reviewed and validated the implementation against approved requirements and test evidence. This document does not attribute software developer, ML engineer or data scientist ownership to the Technical Business Analyst.

Contribution area	Evidence of contribution
Requirements	Converted broad market-research questions into prioritised, testable requirements.
Process	Defined current and future analytical processes and user journeys.
Data	Specified entities, relationships, validation, lineage, KPI grain and privacy rules.
APIs	Defined routes, response purpose, business rules, errors, performance and traceability.
ML	Defined target, eligibility, leakage controls, metrics, explainability, fairness and governance gates.
Governance	Separated implementation, generation, modelling, challenge-label and release authorities.
Testing	Used automated, API, reconciliation, performance, privacy, build and recovery evidence to assess acceptance.
Documentation	Produced stakeholder-ready contracts, decisions, release evidence and handover structure.

28. Portfolio and Professional Summary

28.1 One-page recruiter summary

InsureScope is an insurance market research and predictive analytics platform built around fictional customer, quotation, policy and market data. As Technical Business Analyst, I translated business questions into a governed product scope covering executive, category, product, regional, purchase-channel and customer-segment analysis. I defined KPI rules, data relationships, read-only API requirements, acceptance criteria, NFRs, risk controls and release evidence.

A key technical BA decision was ensuring quotation measures used the quotation region rather than a customer current region, while policy and premium measures followed the originating quotation. I also defined the quotation-to-sale predictive use case, including target labels, feature exclusions, leakage controls, explainability, fairness, reproducibility and human oversight. Prediction remained disabled because no final model was approved.

I challenged a data-governance weakness in which public deterministic SHA-256 selection could reveal confidential challenge Pending membership. The approved solution used a custodian-held secret derivation value outside Git, with public commitment hashes and counts for auditability. I validated delivery using API, database, privacy, reconciliation, build, lint, performance and recovery evidence, and approved the final market-research release candidate with one disclosed performance limitation.

28.2 Five strongest Technical BA achievements

1. Defined and traced the complete insurance market-research scope from business objectives through data rules, APIs, dashboard components and test evidence.
2. Established consistent quotation, conversion, premium, policy and regional attribution rules that prevented misleading KPI calculations.
3. Specified the quotation-to-sale ML use case with target, leakage, explainability, fairness, reproducibility and human-oversight controls.
4. Identified and resolved the challenge-membership confidentiality weakness through a custodian-controlled secret derivation approach.
5. Led evidence-based UAT and release governance, accepting a clear prototype limitation rather than presenting an unsupported performance pass.

28.3 Final CV bullet points

- Defined end-to-end requirements for an insurance market-research platform, linking business objectives, KPI rules, SQL data relationships, eight read-only APIs, dashboard acceptance criteria and release evidence across product, category, region, channel and customer-segment analysis.
- Specified the quotation-to-sale predictive-ML capability, including binary target rules, feature and leakage controls, explainability, fairness, reproducibility, challenge-data confidentiality and human oversight, while maintaining governance that kept unapproved prediction outputs disabled.
- Validated MVP delivery through API, database, privacy, reconciliation, build, lint, performance and recovery evidence; reconciled 10,000 quotations and 5,077 accepted outcomes across six analytical views and documented the customer-segment latency as an accepted prototype limitation.

28.4 Key challenges and resolutions

Challenge	Resolution
Regional KPIs could use an incorrect customer location	Defined quotation-region attribution and policy/premium linkage through the originating quotation.
MVP wording implied prediction that was not implemented	Corrected README and shared notice; verified no predictive route, endpoint or score.
Public deterministic challenge selection could reveal membership	Approved a custodian-held secret derivation value outside Git.
Segment API missed performance target	Accepted as a disclosed non-blocking limitation and prioritised post-release optimisation.

28.5 Tools and skills demonstrated

Business analysis, requirements engineering, insurance domain analysis, process modelling, user journeys, business rules, SQL data requirements, API contracts, NFRs, predictive-ML requirements, explainability, governance, RAID, RTM, UAT, release management, React/Vite, FastAPI/Python, SQL Server, Pandas, scikit-learn, Recharts, pytest, ESLint, Git and GitHub.

29. Evidence Gaps and Unsupported Claims Register

ID	Gap / unsupported claim	Evidence status	Treatment
GAP-01	Final database re-count for 29 tables, 45 foreign keys, 33 checks and 28 indexes	Earlier baseline only	Retain as earlier baseline; perform scripted final schema inventory before external audit.
GAP-02	Final reference counts for 12 regions, 8 segments, 33 products, 66 features, 40 add-ons and 5 competitors	Not repeated in BA-ML-31	Do not present as final re-count without database evidence.
GAP-03	Approved final ML model and metrics	No model approved	Do not report accuracy, precision, recall, F1 or ROC-AUC results.
GAP-04	Frontend automated test count	No frontend test script configured	Add Vitest/RTL and browser tests before claiming automated frontend coverage.
GAP-05	100-user capacity and 500-user load	Not proven	Run realistic concurrent user journeys and database saturation tests.
GAP-06	Production security, privacy and compliance	Not certified	Complete identity, threat, vulnerability, penetration, legal and regulatory reviews.
GAP-07	Measured financial or conversion benefit	Evidence not available	Do not claim savings or uplift; define future measurement plan.
GAP-08	Public application deployment	Not completed at release approval	Treat GitHub as source backup only until hosting is configured and tested.
GAP-09	Power BI in final released web MVP	Evidence not available	Describe as project technology area only, not a verified release component.
GAP-10	Browser accessibility certification	Static checks only	Perform keyboard, contrast, screen-

ID	Gap / unsupported claim	Evidence status	Treatment
			reader and responsive viewport testing.

30. Appendices

Appendix A - Git evidence

Commit	Evidence
bab9d56	ML requirements
4b6326b	Dataset V1 evidence
4cf55fa	V1 modelling evidence; no approved final model
386a898	Dataset V2 evidence
9ba15cd	V2 Gate 2 approval
d865df9	V2 model evidence; no approved final model
d7b4ede	V3 requirements
50f9563	V3 Gate 1 approval
80ef387	V3 pilot 1.2.0 evidence
01c007b	V3 Gate 2 final approval
6840485	BA-ML-27 generation implementation contract
e047e7a	BA-ML-28 confidentiality amendment
4f48398	BA-ML-29 feature schema contract
5cf688d	Feature-contract implementation
238d45d	Temporary feature-row implementation
c5e9a80	BA-ML-30 preprocessing approval
30c2715	Temporary preprocessing implementation
a8aeafc	Prepare Market Research MVP release
ea26f2e	Record Market Research MVP release approval

Appendix B - Complete requirement index

This document contains 36 functional requirements and 24 non-functional requirements, for a total of 60 requirements. Requirement details appear in Sections 9 and 10; traceability appears in Section 21.

Appendix C - Test evidence summary

The document references 27 named test and release-evidence items. The backend automated suite executed 110 tests: 109 passed and one failed on the accepted PERF-001 response-time criterion. Additional lint, build, dependency, route, API, recovery and static accessibility checks are separately identified.

Appendix D - Source hierarchy and conflict resolution

6. Latest approved project document.
7. Latest recorded governance decision.
8. Latest tested implementation.
9. Latest recorded project decision.
10. Older planning documents.

Where older plans described prediction, forecasting, filters, reports, production controls or broader analytical functions, the final release evidence takes precedence: these functions were disabled, absent or recommended rather than represented as implemented.